

// VTX-OS · Architecture Reference · v2.0

System Architecture & Data Flow Diagrams

Visual reference for VTX-OS – from email trigger to Sage 50 journal entry. Six diagrams covering every layer of the system.

GCP · BigQuery

Sage 50 Native SDK

6-Agent Pipeline

Gmail OAuth 2.0

Secret Manager

Double-Entry Bookkeeping

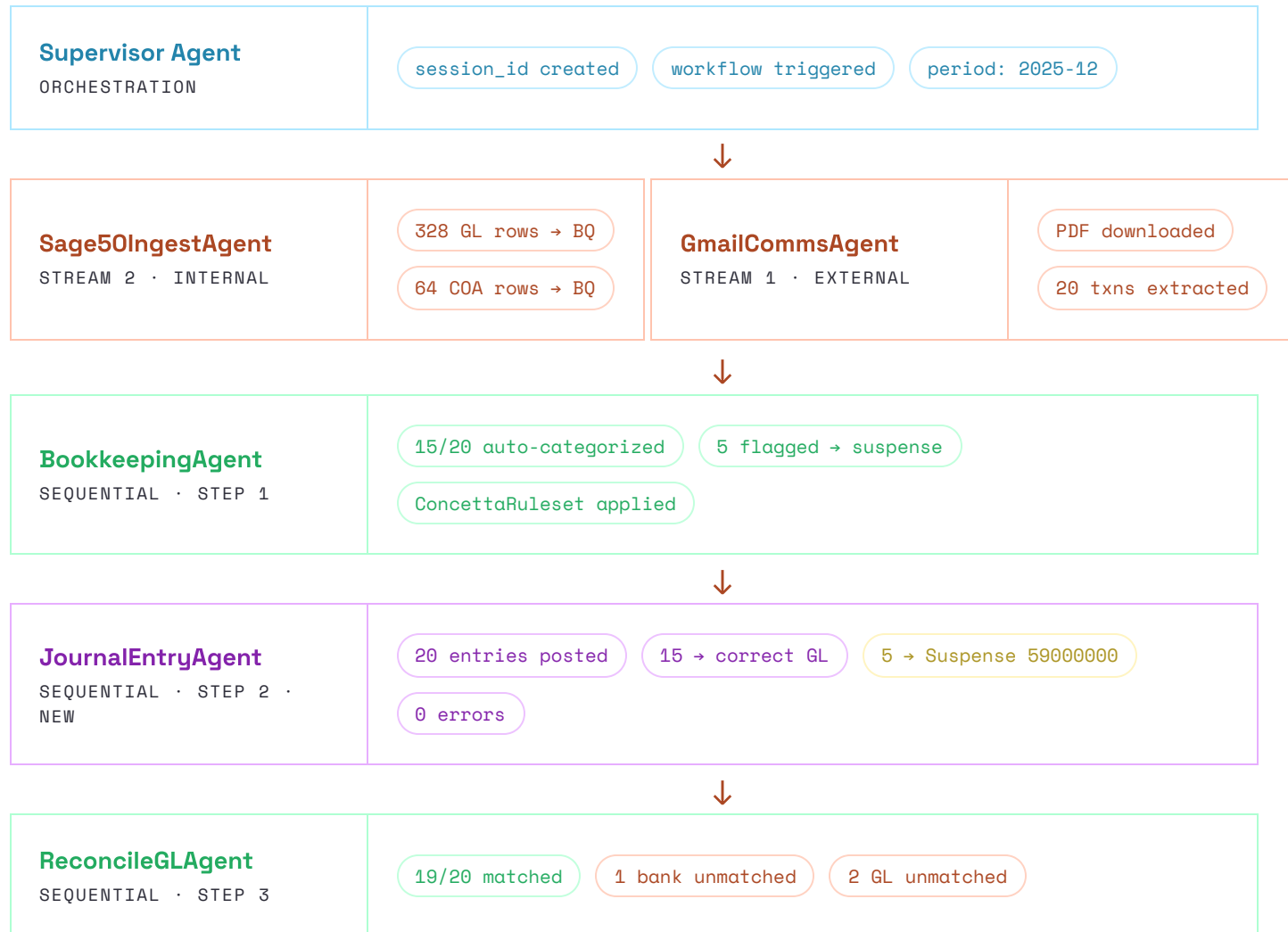
01 System Layers Overview

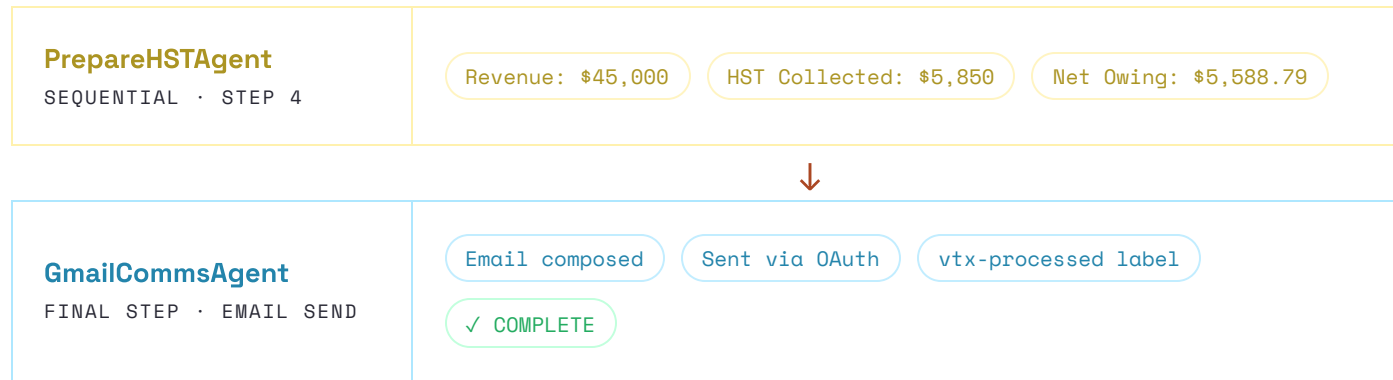
// Six-layer architecture from user interface to data storage



02 Agent Pipeline & Dependencies

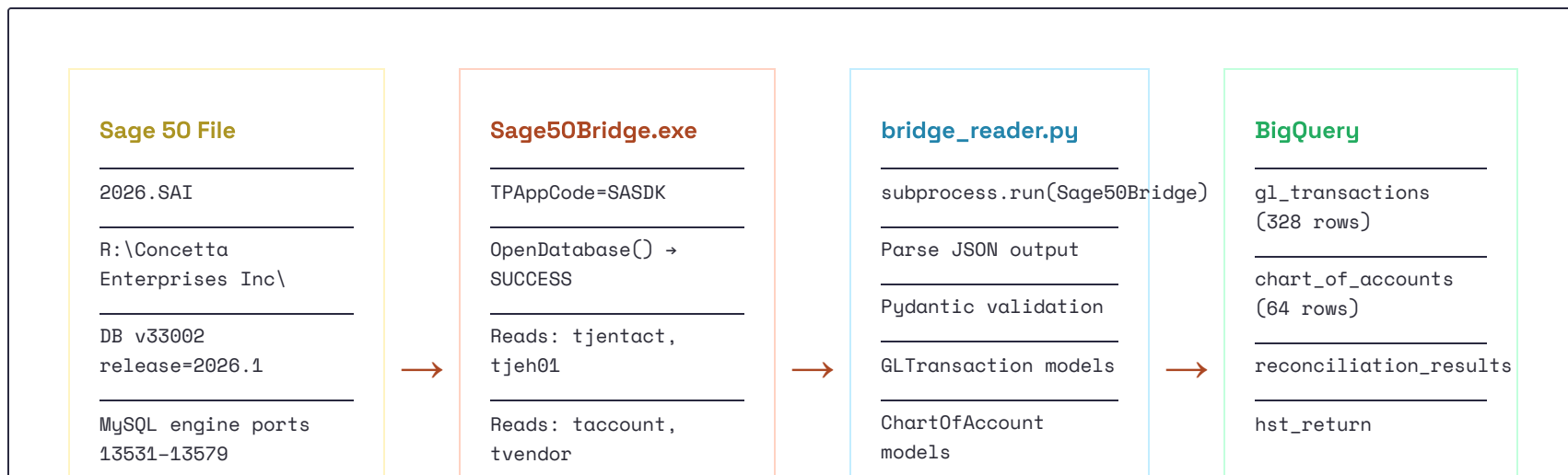
// Sequential execution with dependency enforcement – critical path highlighted





03 Sage 50 Data Flow

// From SAI file through SDK bridge to BigQuery tables



Writes: via
GeneralJournal SDK

Output: JSON → stdout

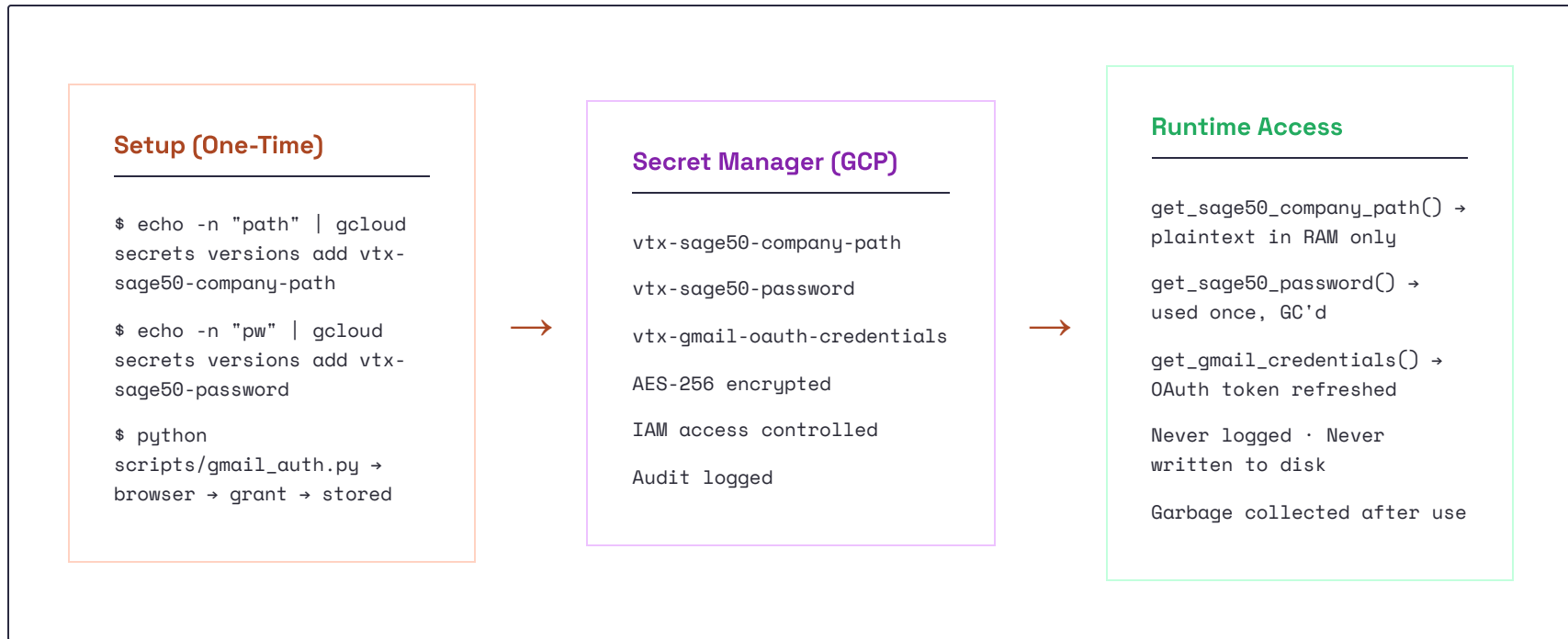
Decimal precision

Audit trail: every
session

vtx-accounting-os-
prod

04 Credential Security Flow

// Zero credentials on disk – all fetched from Secret Manager at runtime



05 Journal Entry Write-Back

// Double-entry bookkeeping rules – how bank transactions become Sage 50 journal entries

DATE	DESCRIPTION	TYPE	DEBIT ACCOUNT	CREDIT ACCOUNT	AMOUNT	STATUS
2025-12-01	SENTRIX FINANCI	Deposit	10600000 Bank	40500000 Revenue	\$23,249.07	✓ Posted

DATE	DESCRIPTION	TYPE	DEBIT ACCOUNT	CREDIT ACCOUNT	AMOUNT	STATUS
2025-12-04	AMEX CARDS	Withdrawal	57250000 AMEX	10600000 Bank	\$450.00	✓ Posted
2025-12-09	SCOTIA VISA	Withdrawal	57000000 Visa	10600000 Bank	\$1,200.00	✓ Posted
2025-12-15	BANK CHARGES	Withdrawal	52000000 Bank Fees	10600000 Bank	\$24.50	✓ Posted
2025-12-01	CHQ-195	Withdrawal	59000000 Suspense	10600000 Bank	\$750.00	⚠ Review
2025-12-03	CASH WITHDRAWAL	Withdrawal	59000000 Suspense	10600000 Bank	\$5,000.00	⚠ Review
TOTALS					All balanced	20 / 0 errors

```
// SDK Write Pattern per entry
```

```
journal = sdk.OpenGeneralJournal() → BeginTransaction() → SetJournalDate("MM/DD/YYYY") → SetAccount(1Id_8digit) → SetDebit(amount) → SetCredit(amount) → SetComment(description[:39]) → Post() → Commit() · RollbackOnError()
```

06 Email Trigger to Completion Timeline

// Full end-to-end sequence from bank statement email to Sage 50 journal posted

T+0:00	Email Arrives dec-2025-bank.pdf received at jquinonez2980@gmail.com · Gmail Watcher polls every 5 seconds · Unread + PDF attachment detected
T+0:02	Period Extracted & PDF Downloaded Filename pattern matched: dec-2025 → period 2025-12 · PDF downloaded (1,049 bytes) · CSV extracted from PDF
T+0:03	Parallel Streams Start Stream 1: 20 bank transactions parsed from CSV · Stream 2: Sage50Bridge opens 2026.SAI (SDK exclusive) · 328 GL + 64 COA rows extracted → BigQuery
T+0:06	Bookkeeping Agent ConcettaRuleset applied · 15/20 transactions matched by keyword + account + amount threshold · 5 CHQ/CASH flagged as ambiguous
T+0:09	Journal Entries Posted to Sage 50 Sage50Bridge opens GeneralJournal via SDK · 20 double-entry pairs created · BeginTransaction → Post → Commit · 15 direct accounts + 5 to Suspense 59000000 · 0 errors
T+0:11	Reconciliation & HST Bank vs GL matching: 19/20 · 1 bank unmatched, 2 GL unmatched · HST: Revenue \$45,000 → Collected \$5,850 → ITCs \$261.21 → Net owing \$5,588.79 · Due 2026-01-31
T+0:14	Email Sent · Complete Professional summary composed · Sent via Gmail OAuth · Subject: "Concetta Enterprises -- 2025-12 Monthly Close" · Original email labelled vtx-processed and marked read · Total elapsed: ~14 seconds

